SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
ACADEMIC HOMEWORK & ASSESSMENT SUBMISSION

Student Name:	RAM CHARAN VANGA
Register No:	RA2511027010164
Course Code & Title:	NPTEL-CS06 - PROGRAMMING, DATA STRUCTURES AND ALGORITHMS USING PYTHON
Task & Channel:	NPTEL Week 5 Assignment: Heaps & Priority Queues [NPTEL]

Assignment Description:

Weekly graded assessment for NPTEL DSA course: 5 MCQs on Min-Heap, Max-Heap, Heapify complexity, and Median maintenance.

Technical Solution:

NPTEL Week 5 Assessment Solutions

Q1: What is the time complexity to build a heap of n elements from an unsorted array?
- **Correct Answer:** `O(n)`
- **Mathematical Rationale:** Building a heap bottom-up via Floyd's build-heap algorithm takes linear time O(n) because the sum of heights sum(h/2^h) converges to a constant.

Q2: In a max-heap with 10 elements, what is the minimum number of comparisons needed to find the minimum element?
- **Correct Answer:** `5`
- **Mathematical Rationale:** In a max-heap of 10 elements, the minimum element must reside in one of the leaf nodes (indices floor(n/2) to n-1, i.e., indices 5 to 9 = 5 leaves). Finding the minimum among 5 leaves requires 4 comparisons.

Submitted by: RAM CHARAN VANGA (RA2511027010164)
Department of Computer Science & Engineering (Big Data Analytics)